

Password Attacks: Cracking Protected Archives

Objective

The objective of this lab is to crack the password of a protected ZIP archive, demonstrating techniques to bypass encryption.

Steps

1. Converted ZIP to Hash

The first step involved converting the ZIP archive into a hash format, making it compatible with password-cracking tools.

```
[eu-academy-6]-[10.10.15.128]-[htb-ac-1246556@htb-f21ouus8qw]-[~/Pentest]
[*]$ zip2john Notes.zip > zip.hash
ver 1.0 efh 5455 efh 7875 Notes.zip/notes.txt PKZIP Encr: 2b chk, TS_chk, cmplen=38, decmplen=26, crc=D0CED23B ts=7EF8 cs=7ef8 type=0
[eu-academy-6]-[10.10.15.128]-[htb-ac-1246556@htb-f21ouus8qw]-[~/Pentest]
[*]$ ls
Notes.zip id_rsa ssh.hash zip.hash
```

2. Mutated Passwords

A list of potential passwords was created using mutation techniques, increasing the chances of finding the correct password.

```
[eu-academy-6]-[10.10.15.128]-[htb-ac-1246556@htb-f21ouus8qw]-[~/Pentest]
[*]$ hashcat --force password.list -r custom.rule --stdout | sort -u > mut_password.list
[eu-academy-6]-[10.10.15.128]-[htb-ac-1246556@htb-f21ouus8qw]-[~/Pentest]
[*]$ ls
Notes.zip Password-Attacks.zip custom.rule id_rsa mut_password.list password.list ssh.hash username.list zip.hash
```

3. Cracked the Password for the ZIP File

Finally, the mutated password list was used to crack the password of the ZIP file, allowing access to its contents.

```
[eu-academy-6]-[10.10.15.128]-[htb-ac-1246556@htb-f21ouus8qw]-[~/Pentest]
[*]$ john --wordlist=mut_password.list zip.hash
Using default input encoding: UTF-8
Loaded 1 password hash (PKZIP [32/64])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
P@ssw0rd3! (Notes.zip/notes.txt)
1g 0:00:00:00 DONE (2024-08-18 06:16) 50.00g/s 2048Kp/s 2048Kc/s 2048KC/s Lovely85!..Patrick
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

Conclusion

This lab illustrated the process of cracking passwords for protected ZIP archives. By converting the archive to a hash, generating mutated passwords, and successfully cracking the password, the lab emphasized the vulnerabilities associated with weak encryption and the importance of strong, unpredictable passwords.